HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LA

LIFE INSURANCE MATHEMATICS A

Semester 1 2022/23

Duration: Two Hours

Total marks: 60

ANSWER ALL 7 QUESTIONS

Electronic calculators approved by the University may be used.

You should be provided with a copy of the yellow 'Formulae and Tables for Actuarial Examinations'.

1. A life office sells two products: endowment insurance policies and term insurance policies. For which of the two would they require more evidence of good health?

Explain your answer.

[Total 2 Marks]

2. Explain whether the following function satisfies the conditions for a survival function of a newborn. If so, state the limiting age for survival.

$$S_0(t) = 1 - \frac{t}{50}.$$

[Total 2 Marks]

3. The basis for this question is:

Mortality: AM92 ultimate
Interest: $i = 12.36\% = (1.06)^2 - 1$ per year effective

Calculate $\ddot{a}_{60}$ to three decimal places.

[Total 4 Marks]

4. For a certain ultimate mortality table, at any integer value of x between 90 and 100 inclusive,

$$q_x = \frac{x - 80}{50}.$$

(a) State if the mortality table is representative of typical human mortality between ages 90 years and 100 years and give a reason for your answer. You may wish to illustrate your reasoning by reference to the values of q_x.

[2 marks]

(b) Calculate the values of

(i) ${}_3p_{95}$. [2 marks]

(ii) ${}_{3|2}q_{91}$. [2 marks]

A select mortality table has a two-year select period, and uses the above ultimate table. For all integer ages x,

$$q_{[x]} = 2q_x \qquad \text{and} \qquad q_{[x]+1} = 1.5q_{x+1}.$$

(c) Calculate ${}_5p_{[93]}$. [2 marks]

(d) Compare the mortality under the select mortality table to the ultimate mortality.

[2 marks]

[Total 10 Marks]

5. A life office sells a one-year term assurance contract. This provides a death benefit of \$500 000 payable at the end of the half-year of death, to a life currently aged exactly 74 years.

The premium basis is as follows:

Mortality: $q_{74} = 0.04164$
Interest: 9% per year effective
Expenses: Nil

(a) Calculate the premium assuming:

(i) A Uniform Distribution of Death (UDD) between integer ages.
[4 marks]

(ii) The force of mortality is constant between integer ages.
[6 marks]

(b) Comment on your answers to part (a)(i) and (a)(ii).
[3 marks]

(c) Without doing any more calculations, explain whether the premium would be higher or lower if the benefit were to be paid immediately on death.
[2 marks]

[Total 15 Marks]

6. The valuation basis for this question is as follows.

Mortality: AM92 ultimate
Interest: 6% per year effective

A life office wishes to sell for £11 700 a life annuity which pays out £1 000 per year to one individual who is aged 60 years. The annuity is paid annually in arrears.

(a) Calculate the expected present value of the profit to the office.

[2 marks]

(b) Calculate the standard deviation of the present value of the profit to the office.

[3 marks]

(c) Calculate the probability that the present value of the life office's profit on this contract will be positive.

[4 marks]

(d) Do you have any concerns about issuing this policy, from a profit viewpoint? Briefly explain your answer.

[2 marks]

(e) Suppose the life annuity was paid annually in advance. Without doing any more calculations, would the standard deviation of the present value of the profit be less than, equal to or greater than the standard deviation calculated in (b)? Explain your answer.

[2 marks]

[Total 13 Marks]

7. A company sells a whole life with-profits policy to a life aged 30. The basic sum assured is £400 000. The death benefit payment is made at the time of death. Premiums are amount £P per annum, with payments made quarterly in advance for at most 35 years.

The pricing basis is:

Mortality:	AM92 Ultimate
Interest:	4% per annum effective
Expenses:	Initial: £300 plus 40% of the first premium
	Renewal: £20 when the client reaches age $31, 32, 33, \ldots$
	Claims: £200 at the time of payment of the sum assured
Bonus:	Simple reversionary bonus of 2.5% per annum vesting at the start of each policy year
Profit:	Expected present value of profit is £3 000

Where appropriate, use Woolhouse's formula or the claim acceleration method.

(a) Write the expected present value of the premiums as a function of P, evaluating the coefficient of P.

[4 marks]

(b) Calculate the expected present value of the benefits including bonus.

[4 marks]

(c) Write down an expression for the expected present value of the expenses. By evaluating the expression, show that it can be reduced to the form $aP + b$, where a and b are constants whose values you must state.

[4 marks]

(d) Hence or otherwise, calculate the annual premium P.

[2 marks]

[Total 14 Marks]

END OF PAPER